EMTL MID – 2 Question Bank Answers

UNIT - III

1. (a) State and explain the poynting theorem and derive the expression for poynting vector.

Ans:

Poynting theorem:

Poynting theorem states that the net power flowing out of a given volume v is equal to the time rate of decrease in the energy stored within v minus the conduction losses.

$$\boldsymbol{P} = \boldsymbol{E} \times \boldsymbol{H} \quad {}^{W}/_{m^2}$$

Proof:

Consider the Maxwell's equations

$$\nabla \times \boldsymbol{E} = -\mu \frac{\partial \boldsymbol{H}}{\partial t} \qquad (1)$$

$$\nabla \times \boldsymbol{H} = \sigma \boldsymbol{E} + \varepsilon \frac{\partial \boldsymbol{E}}{\partial t} \qquad (2)$$

Multiplying equation (2) with **E**, we get

$$\mathbf{E} \cdot \nabla \times \boldsymbol{H} = \sigma \mathbf{E} \cdot \boldsymbol{E} + \varepsilon \mathbf{E} \cdot \frac{\partial \boldsymbol{E}}{\partial t}$$

$$\mathbf{E} \cdot \nabla \times \boldsymbol{H} = \sigma E^2 + \varepsilon \mathbf{E} \cdot \frac{\boldsymbol{\partial E}}{\boldsymbol{\partial t}} \qquad (3)$$

We know that

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = \boldsymbol{H} \cdot (\nabla \times \boldsymbol{E}) - \boldsymbol{E} \cdot (\nabla \times \boldsymbol{H})$$

The above vector identity can be written as

$$\boldsymbol{E} \cdot (\nabla \times \boldsymbol{H}) = \boldsymbol{H}.(\nabla \times \boldsymbol{E}) - \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H})$$

Substitute this in equation (3), we get

$$\boldsymbol{H} \cdot (\nabla \times \boldsymbol{E}) - \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = \sigma E^2 + \varepsilon \mathbf{E} \cdot \frac{\partial \boldsymbol{E}}{\partial t}$$

Substitute equation (1) in above equation, we get

$$-\boldsymbol{H} \cdot \mu \frac{\partial \boldsymbol{H}}{\partial t} - \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = \sigma E^2 + \varepsilon \mathbf{E} \cdot \frac{\partial \boldsymbol{E}}{\partial t}$$

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = -\varepsilon \mathbf{E} \cdot \frac{\partial \boldsymbol{E}}{\partial t} - \mu \boldsymbol{H} \cdot \frac{\partial \boldsymbol{H}}{\partial t} - \sigma E^2 \qquad (4)$$

But

$$\mathbf{E} \cdot \frac{\partial \boldsymbol{E}}{\partial t} = \frac{1}{2} \frac{\partial E^2}{\partial t}$$

$$\mathbf{H} \cdot \frac{\partial \boldsymbol{H}}{\partial t} = \frac{1}{2} \frac{\partial H^2}{\partial t}$$

Substitute these equations in equation (4), we get

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = -\varepsilon \frac{1}{2} \frac{\partial E^2}{\partial t} - \mu \frac{1}{2} \frac{\partial H^2}{\partial t} - \sigma E^2$$

$$\nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) = -\frac{\partial}{\partial t} \left[\frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2 \right] - \sigma E^2$$

Take the volume integral on both sides of above equation, we get

$$\int_V \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) dV = -\int_V \frac{\partial}{\partial t}\left[\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2\right] dV - \int_V \sigma E^2 dV$$

Interchange integration and differentiation, we get

$$\int_V \nabla \cdot (\boldsymbol{E} \times \boldsymbol{H}) dV = -\frac{\partial}{\partial t}\int_V \left[\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2\right] dV - \int_V \sigma E^2 dV$$

Applying the divergence theorem to the left-hand side gives

$$\underset{\text{(I)}}{\oint_S (\boldsymbol{E} \times \boldsymbol{H}) dS} = \underset{\text{(II)}}{-\frac{\partial}{\partial t}\int_V \left[\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2\right] dV} - \underset{\text{(III)}}{\int_V \sigma E^2 dV}$$

Total power leaving the volume	=	Rate of decrease in energy stored in electric and magnetic fields	—	Ohmic power dissipated

$\boldsymbol{E} \times \boldsymbol{H}$ represents the power flow per unit area (w/m^2) and is known as pointing vector.

$$\therefore\ \boldsymbol{P} = \boldsymbol{E} \times \boldsymbol{H}$$

1. (b) Derive the wave equation for uniform plane wave in terms of E.

Ans:

<u>Uniform plane wave:</u>

An EM wave propagating in z - direction is said to be a uniform plane wave if its electric and magnetic fields are independent of x and y – directions.

The following are the properties of uniform plane wave

1. A uniform plane wave propagating in z – direction has no z – component of electric and magnetic fields that means $E_z = 0$ and $H_z = 0$.
2. The electric and magnetic fields of EM wave are always perpendicular to each other.
3. The electric and magnetic fields are independent of x and y – directions therefore

$$\frac{\partial^2 \boldsymbol{E}}{\partial x^2} = 0 \ and\ \frac{\partial^2 \boldsymbol{H}}{\partial x^2} = 0$$

$$\frac{\partial^2 \boldsymbol{E}}{\partial y^2} = 0 \ and\ \frac{\partial^2 \boldsymbol{H}}{\partial y^2} = 0$$

4. The directions of electric and magnetic fields are perpendicular to direction of wave propagation.

<u>Proof:</u>

The uniform plane wave equation in free space is given by

$$\nabla^2 \boldsymbol{E} = \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} \qquad (1)$$

$$\frac{\partial^2 \boldsymbol{E}}{\partial x^2} + \frac{\partial^2 \boldsymbol{E}}{\partial y^2} + \frac{\partial^2 \boldsymbol{E}}{\partial z^2} = \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2}$$

As per definition of uniform plane wave, the electric and magnetic fields are independent of x and y – directions, therefore

$$\frac{\partial^2 \boldsymbol{E}}{\partial x^2} = 0 \; and \; \frac{\partial^2 \boldsymbol{E}}{\partial y^2} = 0$$

Hence equation (1) becomes

$$\frac{\partial^2 \boldsymbol{E}}{\partial z^2} = \mu_0 \varepsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} \qquad (2)$$

Where

$$\boldsymbol{E} = E_x \bar{a}_x + E_y \bar{a}_y + E_z \bar{a}_z$$

Equating the respective components on both sides of equation (2), we get

$$\frac{\partial^2 E_x}{\partial z^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E_x}{\partial t^2} \qquad (3)$$

$$\frac{\partial^2 E_y}{\partial z^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E_y}{\partial t^2} \qquad (4)$$

$$\frac{\partial^2 E_z}{\partial z^2} = \mu_0 \varepsilon_0 \frac{\partial^2 E_z}{\partial t^2} \qquad (5)$$

In free space

$$\nabla . E = 0$$

Therefore

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0$$

Electric field is independent of x and y – directions, therefore

$$\frac{\partial E_z}{\partial z} = 0 \qquad (6)$$

Substitute equation (6) in equation (5), we get

$$\frac{\partial^2 E_z}{\partial t^2} = 0$$

This means that E_z should have one of the following solutions

1. $E_z = 0$
2. $E_z = k$ (constant with time)
3. $E_z = kt$ (i.e increases uniformly with time)

The solutions 2 and 3 will not be a part of wave motion

Therefore $E_z = 0$

Similarly $H_z = 0$

This means that the components of electric and magnetic fields of uniform plane wave in the direction of propagation are zero.

2. (a) Show that $\frac{E}{H} = \sqrt{\frac{\mu}{\varepsilon}}$ for a uniform plane wave.

Ans:

The relation between **E** and **H** is

$$\left|\frac{\boldsymbol{E}}{\boldsymbol{H}}\right| = 120\pi\Omega \approx 377\Omega$$

Proof:

The two Maxwell's equations for free space are given by

$$\nabla \times \boldsymbol{E} = -\mu_0 \frac{\boldsymbol{\partial H}}{\boldsymbol{\partial t}} \qquad (1)$$

$$\nabla \times \boldsymbol{H} = \varepsilon_0 \frac{\boldsymbol{\partial E}}{\boldsymbol{\partial t}} \qquad (2)$$

Expand LHS of equation (2)

$$\begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \varepsilon_0 \frac{\boldsymbol{\partial E}}{\boldsymbol{\partial t}}$$

For a uniform plane wave traveling in the Z-direction, the electric and magnetic fields are independent of x and y – directions, and also electric and magnetic fields have no z component. Hence the above becomes

$$-\frac{\partial H_y}{\partial z}\bar{a}_x + \frac{\partial H_x}{\partial z}\bar{a}_y = \varepsilon_0 \frac{\partial E_x}{\partial t}\bar{a}_x + \varepsilon_0 \frac{\partial E_y}{\partial t}\bar{a}_y \qquad (3)$$

Similarly from equation (1), we get

$$-\frac{\partial E_y}{\partial z}\bar{a}_x + \frac{\partial E_x}{\partial z}\bar{a}_y = -\mu_0 \frac{\partial H_x}{\partial t}\bar{a}_x - \mu_0 \frac{\partial H_y}{\partial t}\bar{a}_y \qquad (4)$$

Equating the respective components, we get

$$-\frac{\partial H_y}{\partial z} = \varepsilon_0 \frac{\partial E_x}{\partial t} \qquad (5)$$

$$\frac{\partial H_x}{\partial z} = \varepsilon_0 \frac{\partial E_y}{\partial t} \qquad (6)$$

$$\frac{\partial E_y}{\partial z} = \mu_0 \frac{\partial H_x}{\partial t} \qquad (7)$$

$$\frac{\partial E_x}{\partial z} = -\mu_0 \frac{\partial H_y}{\partial t} \qquad (8)$$

Now if $E_x = f(z - v_0 t)$

Where $v_0 = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

$$\frac{\partial E_x}{\partial t} = -v_0 f^{|}(z - v_0 t) \qquad (9)$$

Substituting equation (9) in equation (5), we get

$$\frac{\partial H_y}{\partial z} = \varepsilon_0 v_0 f^{|}(z - v_0 t)$$

Integrate the above equation with respect to Z, we get

$$H_y = \sqrt{\frac{\varepsilon_0}{\mu_0}} \int f^{|}(z - v_0 t) dz + K$$

$$H_y = \sqrt{\frac{\varepsilon_0}{\mu_0}} f(z - v_0 t) + K$$

$$H_y = \sqrt{\frac{\varepsilon_0}{\mu_0}} E_x + K$$

As the constant K can't be a part of wave motion, hence we can put K=0

$$H_y = \sqrt{\frac{\varepsilon_0}{\mu_0}} E_x$$

$$\frac{E_x}{H_y} = \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

Similarly if we take $E_y = f(z - v_0 t)$

$$\frac{\partial E_y}{\partial t} = -v_0 f^{|}(z - v_0 t) \qquad (10)$$

Substituting equation (10) in equation (6), we get

$$\frac{\partial H_x}{\partial z} = -\varepsilon_0 v_0 f^{|}(z - v_0 t)$$

Integrate the above equation with respect to Z, we get

$$H_x = -\sqrt{\frac{\varepsilon_0}{\mu_0}} \int f^{|}(z - v_0 t) dz$$

$$H_x = -\sqrt{\frac{\varepsilon_0}{\mu_0}} f(z - v_0 t)$$

$$H_x = -\sqrt{\frac{\varepsilon_0}{\mu_0}} E_y$$

$$\frac{E_y}{H_x} = -\sqrt{\frac{\mu_0}{\varepsilon_0}}$$

$$\frac{E}{H} = \sqrt{\frac{E_x^2 + E_y^2}{H_x^2 + H_y^2}} = \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

$$\frac{E}{H} = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi\Omega = 377\Omega$$

The ratio of electric and magnetic fields is called intrinsic impedance (characteristic impedance) and is denoted by η

$$\eta = \frac{E}{H} = \sqrt{\frac{\mu}{\varepsilon}}$$

For free space

$$\eta = \frac{E}{H} = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi\Omega = 377\Omega$$

2. (b) Explain about the skin depth.

Ans:

<u>Depth of penetration or skin depth:</u>

When a wave propagates through a conducting medium, it is attenuated. Skin depth is defined as the depth at which the wave has been attenuated to 1/e or approximately 37 % of its maximum value. It is denoted by δ.

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

Proof:

Consider a wave propagating through a conducting medium in the z direction.

We know that the amplitude of electric field attenuates approximately exponentially with attenuation constant α. That is,

$$E = E_0 e^{-\alpha z}$$

Where E_0 is maximum amplitude

At a distance equal to skin depth, z = δ, electric field has been attenuated to 1/ e times.

$$\therefore\ E = E_0 e^{-\alpha\delta} = E_0 e^{-1}$$

$$E_0 e^{-\alpha\delta} = E_0 e^{-1}$$

$$e^{-\alpha\delta} = e^{-1}$$

$$\alpha\delta = 1$$

Skin depth is

$$\delta = \frac{1}{\alpha}$$

For a conducting medium

$$\alpha = \sqrt{\frac{\omega\mu\sigma}{2}} = \sqrt{\pi f \mu \sigma}$$

$$\therefore\ \delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

The phenomenon whereby field intensity in a conductor rapidly decreases is known as skin effect. Due to skin effect, the fields are almost confined to a very thin sheet at the surface of the conductor.

3. (a) Discuss the wave propagation in lossless dielectrics.

Ans:

The wave equation in phasor form for perfect dielectric is

$$\nabla^2 \boldsymbol{E} = -\omega^2 \mu\varepsilon\, \boldsymbol{E}$$

$$\frac{\partial^2 \boldsymbol{E}}{\partial x^2} + \frac{\partial^2 \boldsymbol{E}}{\partial y^2} + \frac{\partial^2 \boldsymbol{E}}{\partial z^2} = -\omega^2 \mu\varepsilon\, \boldsymbol{E}$$

For a uniform plane wave the above equation becomes

$$\frac{\partial^2 \boldsymbol{E}}{\partial z^2} = -\beta^2 \boldsymbol{E}$$

Where phase shift constant $\beta = \omega\sqrt{\mu\varepsilon}$

$$\frac{\partial^2 \boldsymbol{E}}{\partial z^2} + \beta^2 \boldsymbol{E} = 0 \qquad (1)$$

Let the electric field be in the x-direction with component E_x, then the equation (1) becomes

$$\frac{\partial^2 E_x}{\partial z^2} + \beta^2 E_x = 0 \qquad (2)$$

The equation (2) is a second order differential equation and the solution of equation (2) is given by

$$E_x(z) = E^+ e^{-j\beta z} + E^- e^{j\beta z} \qquad (3)$$

Where E^+ and E^- are arbitrary constants which can be determined from boundary conditions
The corresponding time-varying field is written as

$$\tilde{E}_x(z,t) = Re\left[E_x(z) e^{j\omega t}\right] \qquad (4)$$

Substitute equation (3) in equation (4), we get

$$\tilde{E}_x(z,t) = Re\left[\left(E^+ e^{-j\beta z} + E^- e^{j\beta z}\right) e^{j\omega t}\right]$$

$$\tilde{E}_x(z,t) = Re\left[E^+ e^{j(\omega t - \beta z)} + E^- e^{j(\omega t + \beta z)}\right]$$

By taking real value

$$\tilde{E}_x(z,t) = E^+ \cos(\omega t - \beta z) + E^- \cos(\omega t + \beta z)$$

The above equation represents the sum of two waves propagating in opposite directions. E^+ represents the electric field component when wave is propagating in the positive z-direction and E^- represents the electric field component when wave is propagating in the negative z-direction.

If the wave is propagating in free space without bounds, it travels in forward direction only and the electric field is given by

$$\tilde{E}_x(z,t) = E^+ \cos(\omega t - \beta z) \quad V/m$$

Similarly, the magnetic field in the y direction is given by

$$\tilde{H}_y(z,t) = H^+ \cos(\omega t - \beta z) + H^- \cos(\omega t + \beta z)$$

If the wave travels only in forward direction,

$$\tilde{H}_y(z,t) = H^+ \cos(\omega t - \beta z) \quad A/m$$

3. (b) Illustrate the concept of EM wave Polarization.

Ans:

<u>Polarization:</u>

The polarization of a uniform plane wave refers to the time-varying behavior of the electric field strength vector at some fixed point in space.
There are three types of polarizations
1. Linear polarization 2. Circular polarization 3. Elliptical polarization

Linear polarization:
Consider a uniform plane wave traveling in the z direction, **E** and **H** vectors lying in x-y plane. If E_y=0 and only E_x is present, the wave is said to be polarized in the x direction. Similarly E_x=0 and only E_y is present, the wave is said to be polarized in the y direction.
If both E_x and E_y are present and are in phase, the resultant electric field makes an angle $\tan^{-1}\left(\frac{E_y}{E_x}\right)$ with x axis.
If the electric field vector is directed always along the line, then the wave is said to linearly polarized. The Fig.1.(a) shows linear polarization.

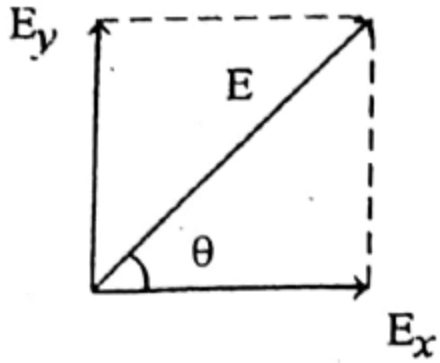


Fig.1.(a) Linear polarization

Circular polarization:
If E_x and E_y have equal magnitudes and a 90°phase difference, the locus of the resultant electric field is a circle and the wave is said to be circularly polarized.

Let E_x and E_y has the equal magnitude Ea and a 90°phase difference between them.
The resultant electric field in vector form is

$$E = E_a\bar{a}_x + jE_a\bar{a}_y$$

The corresponding time varying electric field is

$$\tilde{E}(0,t) = Re\left[Ee^{j\omega t}\right]$$

$$\tilde{E}(0,t) = Re\left[\left(E_a\bar{a}_x + jE_a\bar{a}_y\right)e^{j\omega t}\right]$$

Take real value

$$\tilde{E}(0,t) = E_a\cos\omega t\,\bar{a}_x - E_a\sin\omega t\,\bar{a}_y$$

The corresponding components are

$$E_x = E_a\cos\omega t$$

$$E_y = -E_a\sin\omega t$$

Square and add the above equations, we get

$$E_x^2 + E_y^2 = E_a^2$$

This equation shows that the locus of the resultant electric field traces out a circle of radius E_a and the wave is said to be circularly polarized. It is shown in Fig.1.(b).
If the phase difference between E_x and E_y is + 90°, the rotation of circle is in clockwise direction and the wave is left circularly polarized.
If the phase difference between E_x and E_y is - 90°, the rotation of circle is in counter clockwise direction and the wave is right circularly polarized.

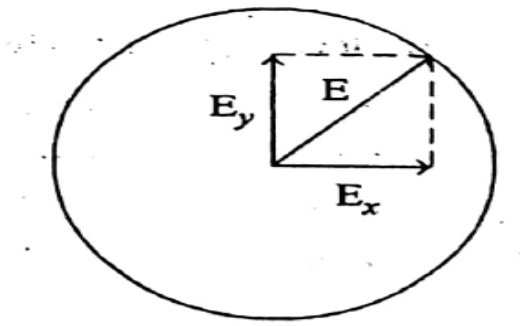


Fig.1.(b) Circular polarization

Elliptical polarization:

If E_x and E_y have different magnitudes and a 90° phase difference, the locus of the resultant electric field is a ellipse and the wave is said to be elliptically polarized.

Let E_x has the magnitude E_a and E_y has the magnitude E_b and a 90° phase difference between them.

The resultant electric field in vector form is

$$E = E_a \bar{a}_x + jE_b \bar{a}_y$$

The corresponding time varying electric field is

$$\tilde{E}(0,t) = Re\left[Ee^{j\omega t}\right]$$

$$\tilde{E}(0,t) = Re\left[\left(E_a \bar{a}_x + jE_b \bar{a}_y\right)e^{j\omega t}\right]$$

Take real value

$$\tilde{E}(0,t) = E_a \cos\omega t\, \bar{a}_x - E_b \sin\omega t\, \bar{a}_y$$

The corresponding components are

$$E_x = E_a \cos\omega t$$

$$E_y = -E_b \sin\omega t$$

Then

$$\frac{E_x}{E_a} = \cos\omega t$$

$$\frac{E_y}{E_b} = -\sin\omega t$$

Square and add the above equations, we get

$$\frac{E_x^2}{E_a^2} + \frac{E_y^2}{E_b^2} = 1$$

This equation shows that the locus of the resultant electric field traces out an ellipse and the wave is said to be elliptically polarized. It is shown in Fig.1.(c).

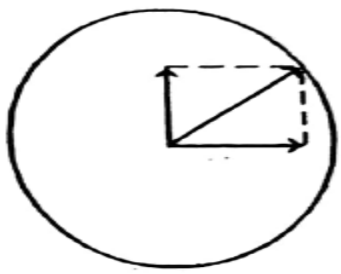


Fig.1.(c) Elliptical polarization

4. (a) Illustrate the concept of reflection of plane wave at normal incidence and hence derive expressions for transmission coefficient and reflection coefficient.

Ans:

Wave incident normally on a perfect dielectric:

When a plane wave is incident on the surface of a perfect dielectric, a part of the energy is transmitted and a part of it is reflected. A perfect dielectric is one with zero conductivity, so that there is no loss or absorption of power in propagating through the dielectric.

Consider two perfect dielectric media separated by a boundary as shown in Fig.1.

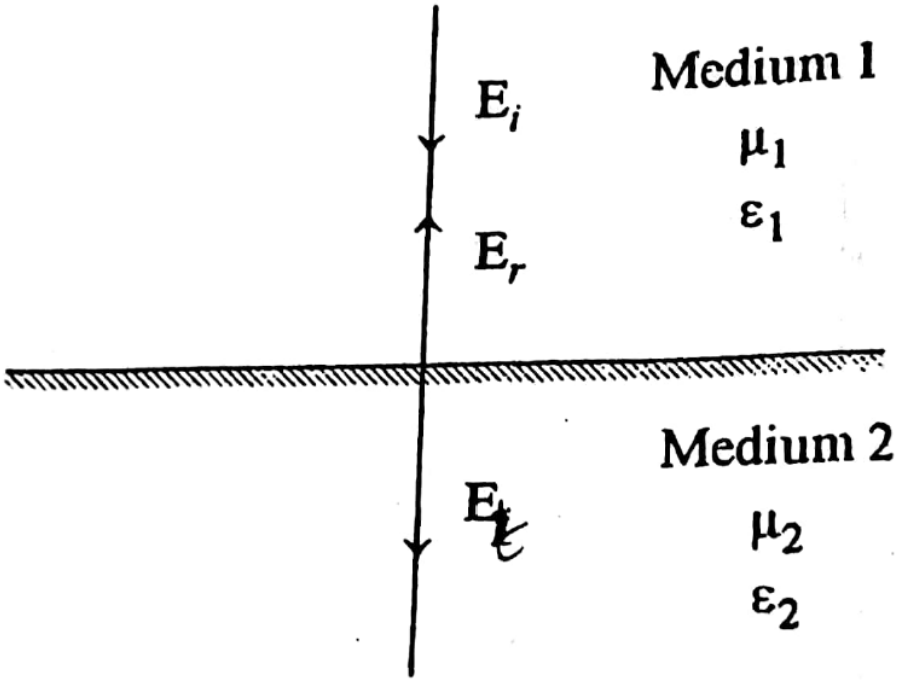


Fig.1. Normal incidence on a perfect dielectric

Let ε_1 and μ_1 are the permittivity and permeability of medium 1 respectively. Let ε_2 and μ_2 are the permittivity and permeability of medium 2 respectively. Let E_i be the electric field intensity of incident wave, E_r be the electric field intensity of reflected wave and E_t be the electric field intensity of transmitted (refracted) wave. Let H_i, H_r and H_t be the magnetic fields of incident, reflected and transmitted waves respectively.

The intrinsic impedance of medium 1 is $\eta_1 = \sqrt{\frac{\mu_1}{\varepsilon_1}}$ and that of medium 2 is $\eta_2 = \sqrt{\frac{\mu_2}{\varepsilon_2}}$

$$E_i = \eta_1 H_i$$
$$E_r = -\eta_1 H_r$$
$$E_t = \eta_2 H_t$$

According to boundary condition, the tangential component of either E or H is continuous across the boundary.

$$H_i + H_r = H_t$$

And

$$E_i + E_r = E_t$$

From above equations

$$H_i = \frac{E_i}{\eta_1}$$
$$H_r = -\frac{E_r}{\eta_1}$$

$$H_t = \frac{E_t}{\eta_2}$$

$$H_t = H_i + H_r = \frac{E_i}{\eta_1} - \frac{E_r}{\eta_1}$$

$$H_t = \frac{1}{\eta_1}(E_i - E_r)$$

$$H_t = \frac{E_t}{\eta_2}$$

$$H_t = \frac{E_i + E_r}{\eta_2}$$

Equating H_t equations

$$\frac{1}{\eta_1}(E_i - E_r) = \frac{1}{\eta_2}(E_i + E_r)$$

$$\eta_2(E_i - E_r) = \eta_1(E_i + E_r)$$

$$\eta_2 E_i - \eta_2 E_r = \eta_1 E_i + \eta_1 E_r$$

$$E_i(\eta_2 - \eta_1) = E_r(\eta_1 + \eta_2)$$

The reflection coefficient

$$\boxed{\frac{E_r}{E_i} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}}$$

The transmission coefficient

$$\frac{E_t}{E_i} = \frac{E_i + E_r}{E_i}$$

$$\frac{E_t}{E_i} = 1 + \frac{E_r}{E_i}$$

$$\frac{E_t}{E_i} = 1 + \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\boxed{\frac{E_t}{E_i} = \frac{2\eta_2}{\eta_2 + \eta_1}}$$

Similarly the reflection coefficient for the magnetic field is given by

$$\frac{H_r}{H_i} = \frac{-\eta_1 E_r}{\eta_1 E_i}$$

$$\frac{H_r}{H_i} = -\frac{E_r}{E_i}$$

$$\frac{H_r}{H_i} = -\left[\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}\right]$$

$$\boxed{\frac{H_r}{H_i} = \frac{\eta_1 - \eta_2}{\eta_1 + \eta_2}}$$

Similarly the transmission coefficient for the magnetic field is given by

$$\frac{H_t}{H_i} = \frac{\eta_1 E_t}{\eta_2 E_i}$$

$$\frac{H_t}{H_i} = \frac{\eta_1}{\eta_2}\frac{E_t}{E_i}$$

$$\frac{H_t}{H_i} = \frac{\eta_1}{\eta_2}\left[\frac{2\eta_2}{\eta_2 + \eta_1}\right]$$

$$\boxed{\frac{H_t}{H_i} = \frac{2\eta_1}{\eta_2 + \eta_1}}$$

The permeabilities of perfect dielectric do not differ from that of free space, hence $\mu_1=\mu_2=\mu_0$. Then

$$\eta_1 = \sqrt{\frac{\mu_0}{\varepsilon_1}} \; and \; \eta_2 = \sqrt{\frac{\mu_0}{\varepsilon_2}}$$

$$\boxed{\frac{E_r}{E_i} = \frac{\sqrt{\varepsilon_1} - \sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2}}}$$

$$\boxed{\frac{E_t}{E_i} = \frac{2\sqrt{\varepsilon_1}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2}}}$$

$$\boxed{\frac{H_r}{H_i} = \frac{\sqrt{\varepsilon_2} - \sqrt{\varepsilon_1}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2}}}$$

$$\boxed{\frac{H_t}{H_i} = \frac{2\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2}}}$$

4. (b) Find the depth of penetration, δ of an EM wave in copper at f = 50 kHz for copper, $\sigma = 5 \times 10^7$ mho/m, $\mu_r = 1$, $€_r = 1$.

Ans:

Given that f = 50 kHz, $\sigma = 5 \times 10^7$ mho/m, $\mu_r = 1$, $€_r = 1$

Depth of penetration, δ of an EM wave is given by

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

$$\delta = \frac{1}{\sqrt{\pi f \mu r \mu 0 \sigma}} = \frac{1}{\sqrt{\pi f \mu 0 \sigma}}$$

$$\delta = \frac{1}{\sqrt{\pi x 50 x 1000 x 4\pi x 10\hat{}-7x5x10\hat{}7}}$$

$$\delta = \frac{1}{\sqrt{\pi\hat{}2x10\hat{}6}}$$

$$\delta = \frac{1}{\pi x 10\hat{}3} = 0.318\, mm$$

UNIT – IV

1. Derive the General solution of Transmission Line and explain its physical significance.

Ans:

<u>Transmission Line Equations:</u> consider a transmission line with two parallel conductors. Let R, L, C and G be the primary parameters. Consider a point P on the line at a distance x from the source as shown in fig.

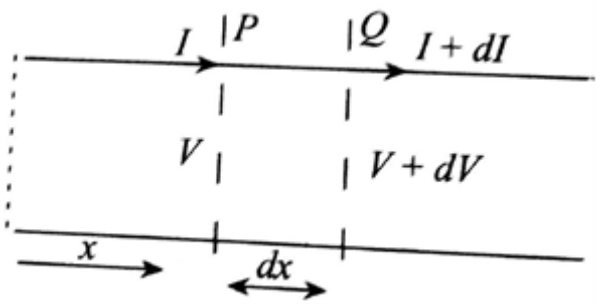


Fig. Voltages and currents on the transmission line

Let Q be the another point at distance dx from the P.

Let V and I be the voltage and current at point P respectively. Let V+dV and I+dI be the voltage and current at point Q respectively.

For a small length dx of the line, the series impedance is $(R + j\omega L)dx$ and shunt admittance is $(G + j\omega C)dx$.

The potential difference between P and Q is

$$V - (V + dV) = I(R + j\omega L)dx$$

$$-\frac{dV}{dx} = (R + j\omega L)I \quad ----(1)$$

The current difference between P and Q is

$$I - (I + dI) = V(G + j\omega C)dx$$

$$-\frac{dI}{dx} = (G + j\omega C)V \quad ----(2)$$

Taking differentiation of eq.(1)

$$-\frac{d^2V}{dx^2} = (R + j\omega L)\frac{dI}{dx} \quad -----(3)$$

Substitute the eq.(2) in eq.(3)

$$\frac{d^2V}{dx^2} = (R + j\omega L)(G + j\omega C)V$$

Similarly differentiate eq.(2) and substitute eq.(1) we get

$$\frac{d^2I}{dx^2} = (R + j\omega L)(G + j\omega C)I$$

The propagation constant

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

Where α=attenuation constant and β=phase shift constant

Therefore

$$\frac{d^2V}{dx^2} = \gamma^2 V \quad ----(4)$$

$$\frac{d^2I}{dx^2} = \gamma^2 I \quad ----(5)$$

The equations (4) and (5) are the second order differential equations in terms of voltage and current whose solutions are given by

$$V = ae^{\gamma x} + be^{-\gamma x} \quad ---(6)$$

$$I = ce^{\gamma x} + de^{-\gamma x} \quad ----(7)$$

Where a, b, c and d are the constants.

In terms of hyperbolic functions the above equations become

$$e^{\gamma x} = \cosh \gamma x + \sinh \gamma x$$

$$e^{-\gamma x} = \cosh \gamma x - \sinh \gamma x$$

$$V = A \cosh \gamma x + B \sinh \gamma x \quad --(8)$$

$$I = C \cosh \gamma x + D \sinh \gamma x \quad --(9)$$

Instead of four constants A,B,C and D the above equations can simplified to only two constants, by substituting the value of V from eq.(8) in eq.(1)

$$-\frac{d}{dx}(A \cosh \gamma x + B \sinh \gamma x) = (R + j\omega L)I$$

$$-(\gamma A \sinh \gamma x + B\gamma \cosh \gamma x) = (R + j\omega L)I$$

$$I = \frac{-\gamma}{(R + j\omega L)}(A \sinh \gamma x + B \cosh \gamma x)$$

$$I = -\sqrt{\frac{G + j\omega C}{R + j\omega L}}(A \sinh \gamma x + B \cosh \gamma x)$$

$$I = -\frac{1}{Z_0}(A \sinh \gamma x + B \cosh \gamma x)$$

Where

$$\text{Characteristic impedance } Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$V = A \cosh \gamma x + B \sinh \gamma x$$

$$I = -\frac{1}{Z_0}(A \sinh \gamma x + B \cosh \gamma x)$$

The constants A and B can be obtained by using initial conditions.

Let V_s and I_s be the source voltage and current respectively. At source end, x=0, the voltage $V= V_s$ and current $I= I_s$

Then

$$V_s = A \cosh \gamma(0) + B \sinh \gamma(0)$$

Therefore $$V_s = A$$

$$I_s = -\frac{1}{Z_0}(A \sinh \gamma(0) + B \cosh \gamma(0))$$

Therefore $$B = -I_s Z_0$$

Substituting the constants A and B in above equations

$$V = V_s \cosh \gamma x - I_s Z_0 \sinh \gamma x$$

$$I = I_s \cosh \gamma x - \frac{V_s}{Z_0} \sinh \gamma x$$

These are called transmission line equations. They give voltage and current at a point of distance x from the sending end in terms of source voltage and current.

2. (a) Derive the attenuation constant α and phase constant β from the propagation constant.

Ans:

Attenuation and phase constants:

The propagation constant γ is a complex quantity.

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

Squaring the magnitude of γ we get

$$\alpha^2 + \beta^2 = \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)} \quad ---(1)$$

Squaring γ on the both sides

$$(\alpha + j\beta)^2 = (R + j\omega L)(G + j\omega C)$$

$$\alpha^2 - \beta^2 + j2\alpha\beta = RG - \omega^2 LC + j(\omega LG + \omega RC)$$

Equating the real parts we get

$$\alpha^2 - \beta^2 = RG - \omega^2 LC \quad ----(2)$$

Adding Eq.(1) and Eq.(2)

$$2\alpha^2 = \sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)} + RG - \omega^2 LC$$

The real part of γ is called attenuation constant. It determines the reduction or attenuation in voltage and current along the line. Its unit is neper per km. 1neper=8.686dB.

$$\alpha = \sqrt{\frac{1}{2}\left[\sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)} + (RG - \omega^2 LC)\right]} \quad NP/km$$

Subtracting Eq.(2) from Eq.(1)

$$\beta = \sqrt{\frac{1}{2}\left[\sqrt{(R^2 + \omega^2 L^2)(G^2 + \omega^2 C^2)} - (RG - \omega^2 LC)\right]} \quad rad/km$$

The imaginary part of γ is called the phase constant. It determines the variation in phase position of voltage and current along the line. Its unit is radians per km.

(b) Derive the characteristic impedance Z_0 of transmission line.

Ans:

Characteristic impedance(Z_0) : The characteristic impedance Z_0 of the line is the ratio of positively
traveling voltage wave to current wave at any point on the line. The characteristic impedance is also
defined as the input impedance of an infinite line.

We know that

$$-\frac{dV}{dx} = (R + j\omega L)I$$

For infinite line

$$V = V_S e^{-\gamma x}$$

$$I = I_S e^{-\gamma x}$$

$$-\frac{d}{dx}(V_S e^{-\gamma x}) = (R + j\omega L) I_S e^{-\gamma x}$$

$$\gamma V_S e^{-\gamma x} = (R + j\omega L) I_S e^{-\gamma x}$$

$$\gamma V_S = (R + j\omega L) I_S$$

$$\frac{V_S}{I_S} = \frac{R + j\omega L}{\gamma}$$

The input impedance of infinite line is given by

$$Z_0 = \frac{V_S}{I_S}$$

$$\text{Characteristic impedance } Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

The propagation constant γ and the characteristic impedance Zo are important properties of the line because they both depend on the line parameters *R*, *L*, *G*, and *C* and the frequency of operation.

3. (a) List the conditions of Lossless transmission line.

Ans:

<u>Lossless Line *(R* = 0 = *G)*:</u> A transmission line is said to be loss less if the conductors of the line
are perfect($\sigma_c \approx \infty$) and the dielectric medium separating them is lossless ($\sigma_d \approx 0$).
Condition for losses less line is, R = 0 = *G*.
The high frequency lines are termed as lossless lines because the $\omega = 2\pi f$ in the series impedance($R + j\omega L$) and shunt admittance ($G + j\omega C$) becomes very large due to the high frequency.
Therefore the real part of the series impedance can be neglected.
The propagation constant and characteristic impedance for losses line are given by

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Substitute R = 0 = *G*

$$\gamma = \alpha + j\beta = j\omega\sqrt{LC}$$

Therefore

$$\alpha = 0 \text{ and } \beta = \omega\sqrt{LC}$$

$$Z_0 = \sqrt{\frac{L}{C}}$$

The phase velocity is

$$v_0 = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$$

3. (b) Illustrate the concept of Equivalent Circuit of transmission line.

Ans:

<u>T-Equivalent Circuit of transmission line:</u>

The series impedance Z and shunt admittance Y of the line per unit length can be expressed as

$$Z = R + j\omega L$$
$$Y = G + j\omega C$$

The T-equivalent circuit of the transmission line is shown in Figure 1.

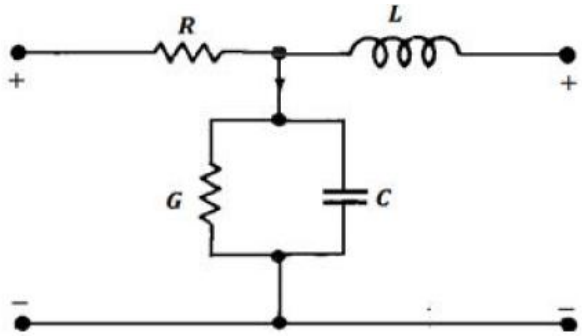


Figure 1: T-Equivalent circuit of transmission line

π -Equivalent Circuit of transmission line:

A " π equivalent circuit" of a transmission line is a simplified electrical circuit representation where the line's distributed capacitance is split in half and placed at each end, with a single series impedance in the middle, creating a circuit shape resembling the Greek letter " π ". This model is commonly used in power system analysis to accurately calculate voltage and current values at different points along the line, especially for medium-length transmission lines. The π-equivalent circuit of the transmission line is shown in Figure 2.

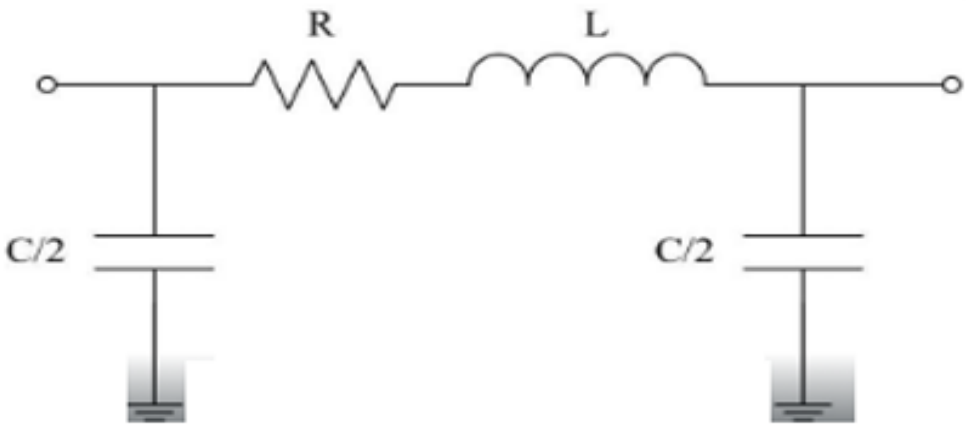


Figure 2: Pi Equivalent circuit of transmission line

4. (a) List the conditions of distortion less transmission line.

Ans:

Distortion less Line*(R/L = G/C):* A transmission line is said to be distortion less if the attenuation
constant 'α' is frequency independent while the phase constant 'β' is linearly dependent on frequency.
The condition for the distortion less line is *R/L = G/C*.
The propagation constant for distortion less line is given by

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{RG\left(1 + \frac{j\omega L}{R}\right)\left(1 + \frac{j\omega C}{G}\right)}$$

For the distortion less line

$$\frac{R}{L} = \frac{G}{C}$$

Therefore

$$\gamma = \sqrt{RG}\left(1 + \frac{j\omega C}{G}\right)$$

$$\alpha + j\beta = \sqrt{RG} + j\omega LC$$

$$\alpha = \sqrt{RG} \quad \text{and} \quad \beta = \omega\sqrt{LC}$$

The characteristic impedance for distortion less line is given by

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{R(1 + j\omega L/R)}{G(1 + j\omega C/G)}} = \sqrt{\frac{R}{G}}$$

$$R_0 = \sqrt{\frac{R}{G}} = \sqrt{\frac{L}{C}} \qquad \text{and } X_0 = 0$$

The phase velocity is

$$v_0 = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$$

4. (b) Distinguish the various types of transmission lines.

Ans:

A transmission line basically consists of two or more parallel conductors used to connect a source to a load. The source may be a generator, a transmitter, or an oscillator; the load may be a factory, an antenna, or an oscilloscope, respectively.
Transmission lines are commonly used in power distribution, communications, electrical laboratories and transmission lines such as the twisted-pair and coaxial cables are used in computer networks such as the ethernet and internet.
Transmission line problems are usually solved using EM field theory and electric circuit theory, the two major theories on which electrical engineering is based.

Types of Transmission lines:

a) Two-wire line: This transmission line consists of a pair of parallel conducting wires separated by
a uniform distance .These are used in power systems or telephones lines.

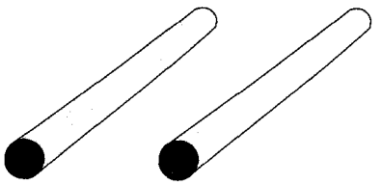


Fig. Two-wire line.

b) Coaxial line: The co-axial cable can be constructed by placing a solid conductor inside a hallow
cylindrical coaxially and these two are separated by a dielectric. They are used as TV cables, telephones cables and power cables.

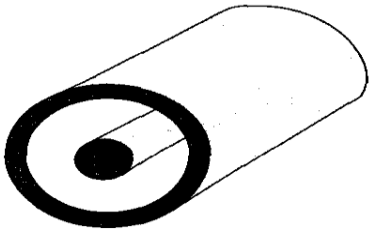
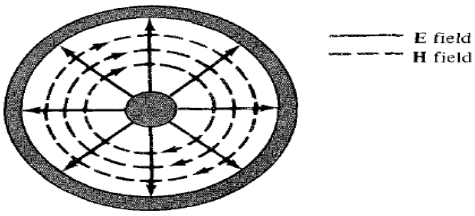
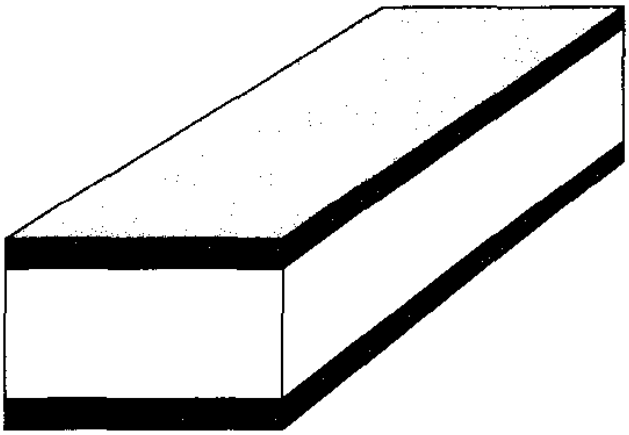


Fig.(a) Coaxial cable

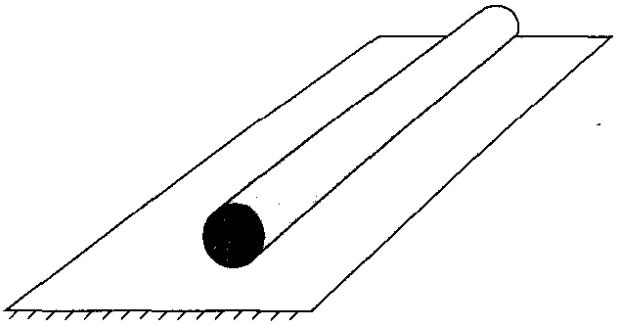


(b).E and H fields in coaxial cable

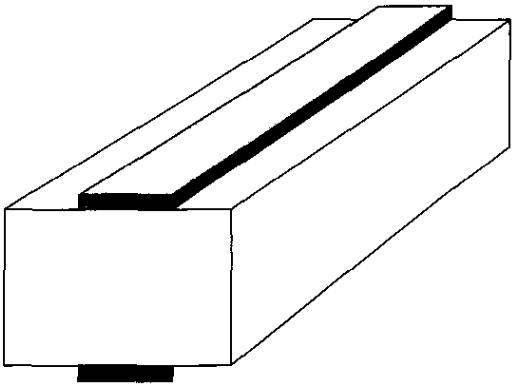
(c) Planar line: It consists of two parallel conductors separated by dielectric medium as shown in fig. It supports TE and TM waves.



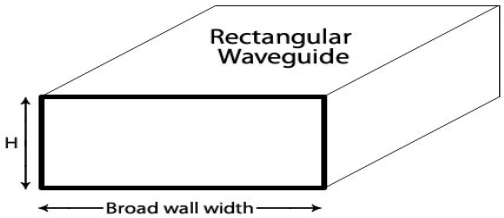
d) Wire above conducting plane: It consists of a conducting wire above the ground plane as shown in fig.



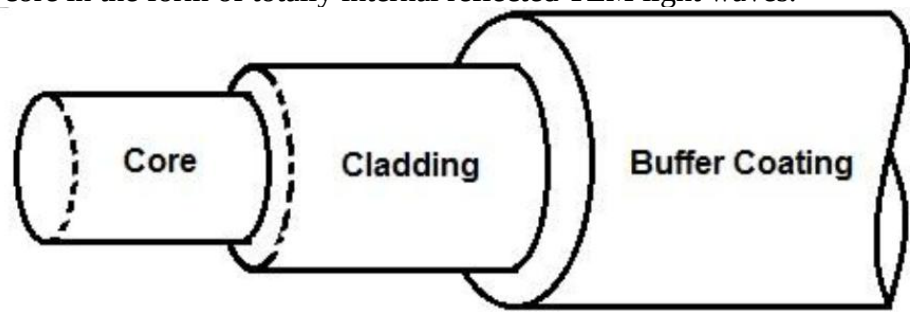
e) Micro strip line: The Micro strip line is transmission line geometry with a single conductor trace on one side of a dielectric substrate and a single ground plane on the other side.



f) Wave guides: The wave guides are hallowed or dielectric filled conductor used to transmit the
electromagnetic energy at micro wave frequency ranges.
The wave propagates in TE ,TM,TEM modes.



g) Optical fibers transmission line: It consists of core and cladding. Information passes through the core in the form of totally internal reflected TEM light waves.



UNIT – V

1. (a) Derive an expression for input impedance at any point on the transmission line.

Ans:

A transmission line terminated with any load impedance Z_R at x=l is shown in fig.

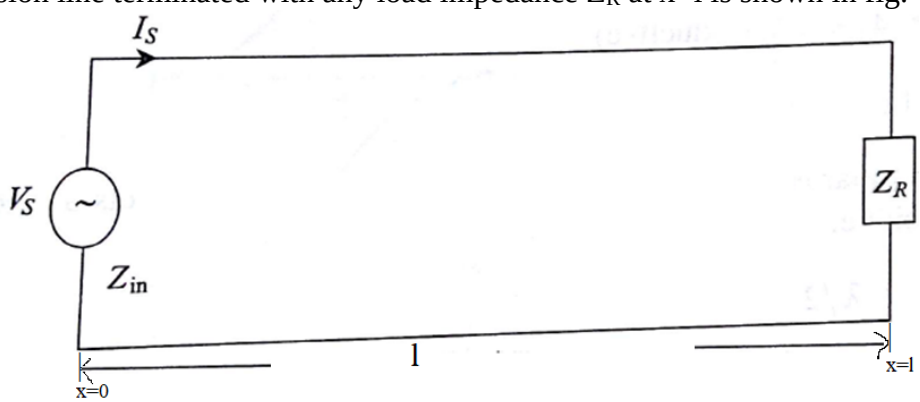


Fig. Input impedance of transmission line

The V and I equations at a point of distance x from the sending end in terms of source voltage and current are given by.

$$V = V_s \cosh \gamma x - I_s Z_0 \sinh \gamma x$$

$$I = I_s \cosh \gamma x - \frac{V_s}{Z_0} \sinh \gamma x$$

The voltage and currents at load end (x=l) are given by

$$V_R = V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l$$

$$I_R = I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l$$

The load impedance Z_R is given by

$$Z_R = \frac{V_R}{I_R}$$

$$Z_R = \frac{V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l}{I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l}$$

$$Z_R \left(I_s \cosh \gamma l - \frac{V_s}{Z_0} \sinh \gamma l \right) = V_s \cosh \gamma l - I_s Z_0 \sinh \gamma l$$

$$Z_R I_s \cosh\gamma l - Z_R \frac{V_s}{Z_0}\sinh\gamma l = V_s \cosh\gamma l - I_s Z_0 \sinh\gamma l$$

$$Z_R I_s \cosh\gamma l + I_s Z_0 \sinh\gamma l = V_s \cosh\gamma l + Z_R \frac{V_s}{Z_0}\sinh\gamma l$$

$$I_s(Z_R \cosh\gamma l + Z_0 \sinh\gamma l) = V_s\left(\cosh\gamma l + \frac{Z_R}{Z_0}\sinh\gamma l\right)$$

$$\frac{V_s}{I_s} = \frac{Z_R \cosh\gamma l + Z_0 \sinh\gamma l}{\cosh\gamma l + \frac{Z_R}{Z_0}\sinh\gamma l}$$

The input impedance of the line is given by

$$Z_{in} = \frac{V_s}{I_s}$$

$$Z_{in} = Z_0\left[\frac{Z_R \cosh\gamma l + Z_0 \sinh\gamma l}{Z_0 \cosh\gamma l + Z_R \sinh\gamma l}\right]$$

$$Z_{in} = Z_0\left[\frac{Z_R + Z_0 \tanh\gamma l}{Z_0 + Z_R \tanh\gamma l}\right]$$

For lossless line γ=jβ

$$Z_{in} = Z_0\left[\frac{Z_R + jZ_0 \tan\beta l}{Z_0 + jZ_R \tan\beta l}\right]$$

1. (b) Illustrate the variation of Z_{in} of lossless transmission line when SC & OC with different lengths.

Ans:

<u>Short circuited line (Z_R = 0):</u> If transmission line is terminated with short circuit, then the line is called short circuited line.

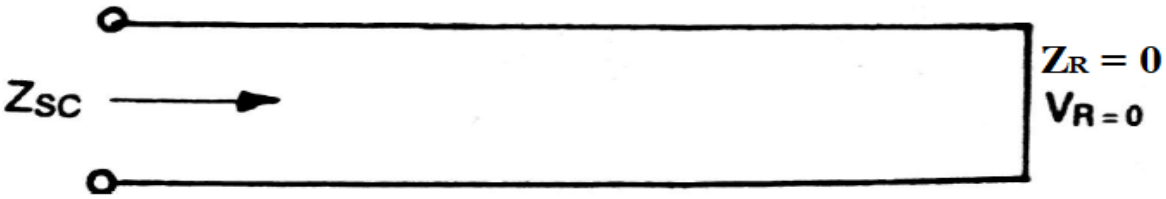


Figure 1. Short circuit transmission line

The input impedance of the short circuited line can be obtained by substituting the $Z_R = 0$ in the input impedance of the transmission line.

$$Z_{in} = Z_0\left[\frac{Z_R + Z_0 \tanh\gamma l}{Z_0 + Z_R \tanh\gamma l}\right]$$

If Z_R=0 then

$$Z_{sc} = Z_0 \tanh\gamma l$$

The input impedance of the short circuited lossless transmission line is given by

$$Z_{sc} = jZ_0 \tan\beta l$$

The reflection coefficient of the short circuited transmission line is K= -1.

The VSWR of the short circuited transmission line is S=∞.

The variation of input impedance of SC line is shown in figure 2.

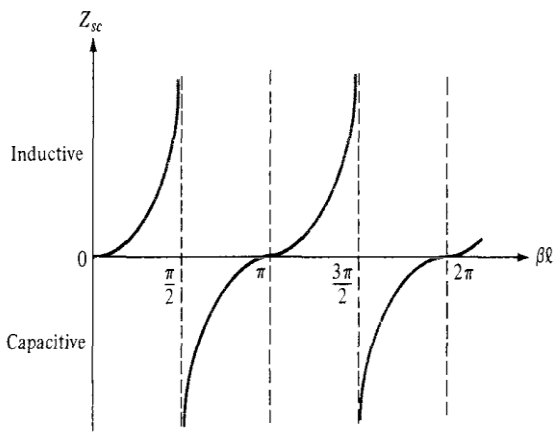


Figure 2. Input impedance of SC

<u>Open circuited line ($Z_R = \infty$):</u> If transmission line is terminated with open circuit, then the line is called open circuited line.

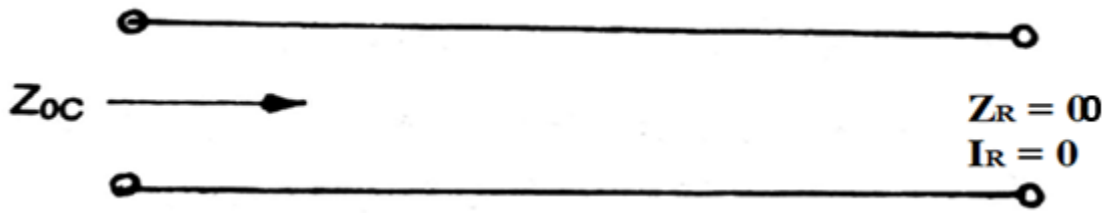


Figure 1. Open circuit transmission line

The input impedance of the open circuited line can be obtained by substituting the $Z_R = \infty$ in the input impedance of the transmission line.

$$Z_{in} = Z_0 \left[\frac{Z_R + Z_0 \tanh \gamma l}{Z_0 + Z_R \tanh \gamma l}\right]$$

If $Z_R = \infty$ then

$$Z_{oc} = Z_0 \coth \gamma l$$

The input impedance of the open circuited line lossless transmission line is given by

$$Z_{oc} = -jZ_0 \cot \beta l$$

The reflection coefficient of the open circuited line transmission line is K= 1. The VSWR of the open circuited line transmission line is S=∞. The variation of input impedance of OC line is shown in Figure 2.

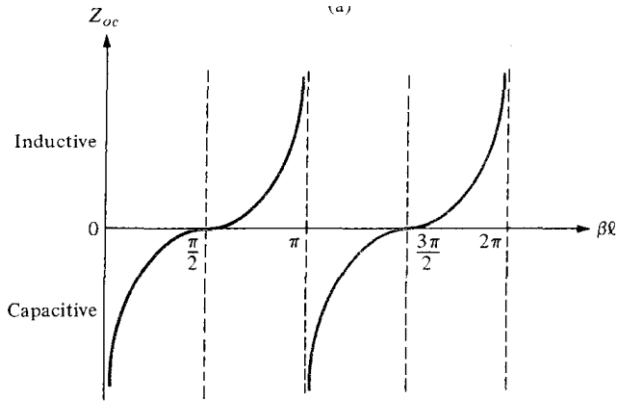


Figure 2. Input impedance of OC

2. (a) List the physical significance of Smith chart and its applications in transmission line.

Ans:

The Smith Chart:

The Smith chart is the most commonly used graphical techniques in solving transmission problems in simple way. It is basically a graphical indication of the impedance and VSWR of a transmission line as one moves along the line.

The construction of the chart is based on the reflection coefficient

$$K = \frac{Z_R - Z_0}{Z_R + Z_0}$$

Instead of having separate Smith charts for transmission lines with different characteristic impedances such as Zo = 60,100, and 120. A normalized chart can be used for any line in which all impedances are normalized with respect to the characteristic impedance Zo of the particular line under consideration.

$$K = \frac{\frac{Z_R}{Z_0} - 1}{\frac{Z_R}{Z_0} + 1} = \frac{z_r - 1}{z_r + 1}$$

Where z_r is the normalized impedance

$$z_r = \frac{1 + K}{1 - K}$$

Since z_r and K both are complex quantities, we have

$$R + jX = \frac{1 + K_r + jK_x}{1 - (K_r + jK_x)}$$

$$R + jX = \frac{(1 + K_r) + jK_x}{(1 - K_r) - jK_x}$$

Rationalizing on the right hand side, we get

$$R + jX = \frac{(1 + K_r) + jK_x}{(1 - K_r) - jK_x} \times \frac{(1 - K_r) + jK_x}{(1 - K_r) + jK_x}$$

$$R + jX = \frac{1 - K_r^2 - K_x^2 + 2jK_x}{(1 - K_r)^2 + K_x^2}$$

Equating real and imaginary parts on both sides, we get

$$R = \frac{1 - K_r^2 - K_x^2}{(1 - K_r)^2 + K_x^2} \quad ----(1)$$

$$X = \frac{2K_x}{(1 - K_r)^2 + K_x^2} \quad ---(2)$$

Eq.s(1) and (2) will yield two set of orthogonal circles when solved separately. Eq.(1) will results in family of circle called R-circle while Eq.(2) will results in family of circle called X-circle.

(i) The constant R-circle:

By simplifying Eq.(1), we get

$$\left(K_r - \frac{R}{1 + R}\right)^2 + K_x^2 = \left(\frac{1}{1 + R}\right)^2$$

This equation represents a family of circles on the reflection co-efficient plane. These circles are called constant –R circles with center $\left(\frac{R}{1+R}, 0\right)$ and radius $\frac{1}{1+R}$.

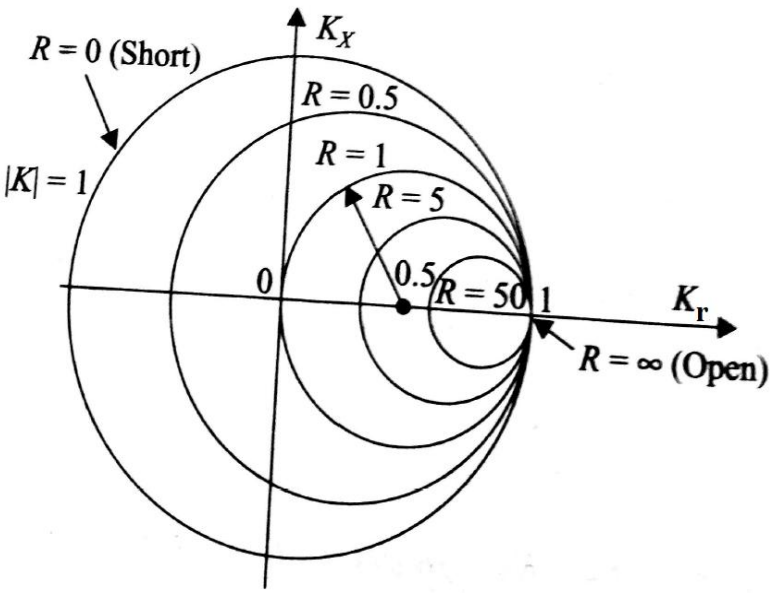


Fig. Family of constant R circles

(ii) The constant X-circle:

By simplifying Eq.(2), we get

$$(K_r - 1)^2 + \left(K_x - \frac{1}{X}\right)^2 = \left(\frac{1}{X}\right)^2$$

This equation represents a family of circles on the reflection co-efficient plane. These circles are called constant –X circles with center $\left(1, \frac{1}{X}\right)$ and radius $\frac{1}{X}$.

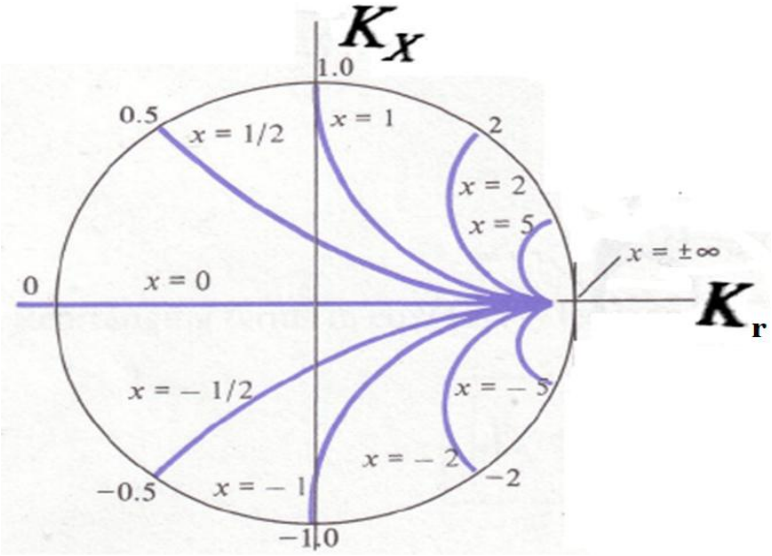


Fig. Family of constant X circles

The complete smith chart can be obtained by the superposition of the family of the constant-R circles and constant-X circles on K plane as shown in fig.

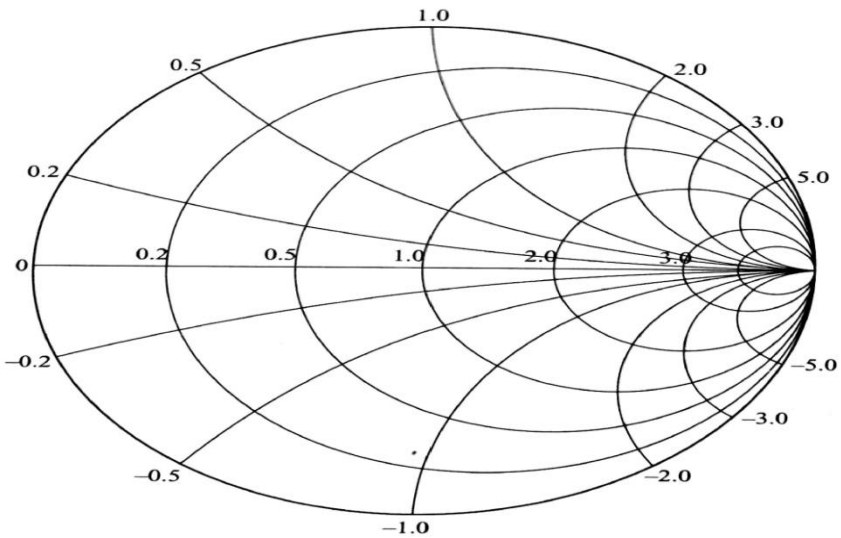


Fig. Smith chart

Properties of smith chart:

1. The smith chart can be used for impedance as well as for admittance.
2. The smith chart consists of constant –R circles and constant –X circles.
3. The point P_{sc} on the chart represents short circuit or zero impedance and the point P_{oc} on the chart
   represents open circuit or infinite impedance.
4. A complete revolution of 360° around the Smith chart represents a distance of λ/2 on the line.
5. Clockwise movement on the chart is regarded as moving toward the generator from the load on the transmission line. Similarly, counterclockwise movement on the chart corresponds to moving toward the load from the generator.
6. The distance between Z_{sc} and Z_{oc} is λ/4.

Applications of smith chart:

1. Plotting impedance
2. Measurement of VSWR
3. Measurement of reflection coefficient K

4. Measurement of input impedance of the transmission line.
5. Impedance to admittance conversion
6. Finding voltage maximum and voltage minimum locations
7. Stub matching

2. (b) For a transmission line which is terminated in a normalized impedance Zn, VSWR = 2. Find the normalized impedance magnitude.

Ans:

Given that VSWR, $S = 2$.

We know that $S = \frac{1+|K|}{1-|K|}$

$$2 = \frac{1+|K|}{1-|K|}$$

$$2(1-|K|) = 1+|K|$$

$$2 - 2|K| = 1 + |K|$$

$$2 - 1 = |K| + 2|K|$$

$$1 = 3|K|$$

$$\Rightarrow \boxed{|K| = \frac{1}{3} = 0.33}$$

Normalized impedance is given by

$$Z_r = \frac{1+K}{1-K}$$

$$= \frac{1+0.33}{1-0.33}$$

$$= \frac{1.33}{0.67}$$

$$= 1.985$$

$$\boxed{Z_r \approx 2}$$

$\therefore$ Magnitude of normalized impedance is 2.

3. (a) Show that infinite line is equivalent to a finite line terminated in Z_0.

Ans:

Transmission line terminated with characteristic impedance:
Consider a line of length *l* which is terminated with characteristic impedance Z_0 as shown in fig.

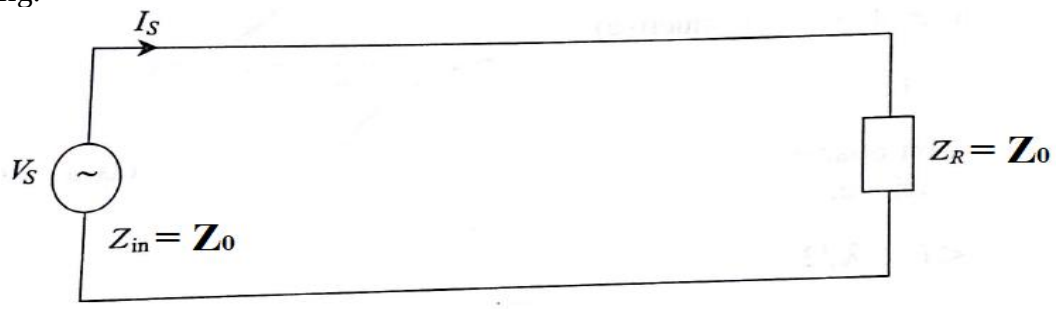


Fig. Transmission line terminated with Z_0

The input impedance of the line is given by

$$Z_{in} = Z_0 \left[\frac{Z_R + Z_0 \tanh \gamma l}{Z_0 + Z_R \tanh \gamma l}\right]$$

Here $Z_R = Z_0$

$$Z_{in} = Z_0 \left[\frac{Z_0 + Z_0 \tanh \gamma l}{Z_0 + Z_0 \tanh \gamma l}\right]$$

$$Z_{in} = Z_0$$

Hence the input impedance of a finite line terminated with characteristic impedance Z_0 is equal to characteristic impedance Z_0.
Therefore a finite line terminated with characteristic impedance Z_0 is equal to infinite line.
A line terminated with characteristic impedance Z_0 is called matched line.

3. (b) Explain principle of impedance matching with quarter wave transformer.

Ans:

When the transmission line is terminated with a load impedance which is not equal to the characteristic impedance of the line, mismatch occurs and reflected wave exists on the line. Mismatch reduces efficiency and increases power loss. Quarter wave line is inserted between the line and load to match load impedance to the line as shown in Fig.

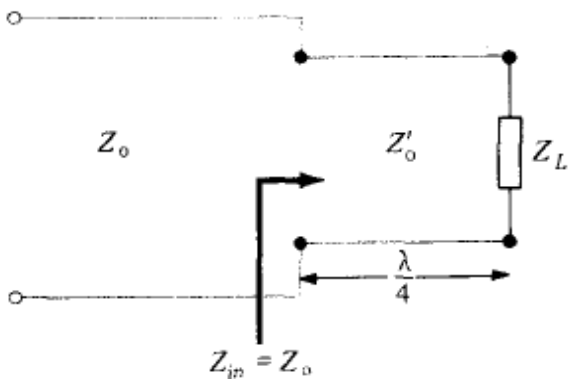


Fig. Quarter wave transformer

Z_0^1 is selected such that $Z_{in} = Z_0$

$$Z_0^l = \sqrt{Z_0 Z_L}$$

The λ/4 line is also called Quarter wave transformer.

Advantages:

1. Simple to design

Disadvantages:

1. We have to cut the line to insert a quarter wave transformer in between the line and the load.
2. It is frequency sensitive.

4. (a) Illustrates how UHF Lines act as Circuit Elements.

Ans:

Different lengths of transmission lines can be used as circuit elements as discussed below.

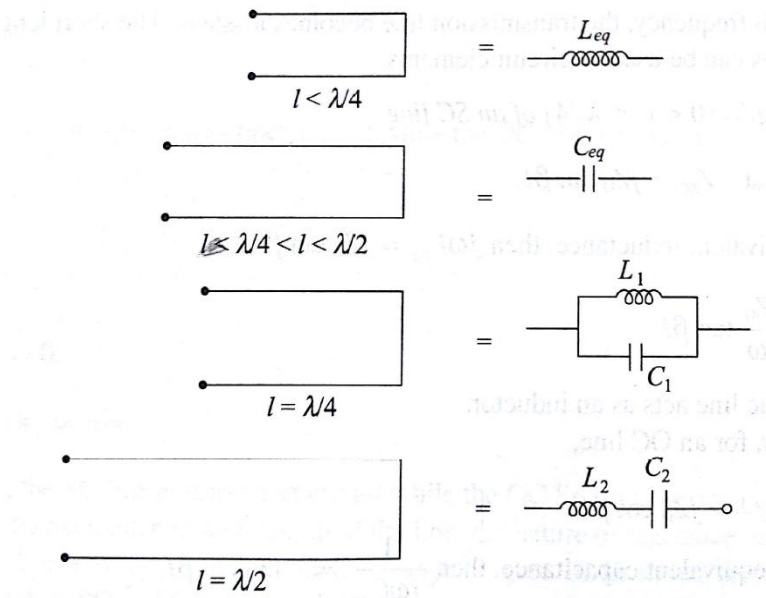


Fig. SC lines at different lengths with equivalent circuits

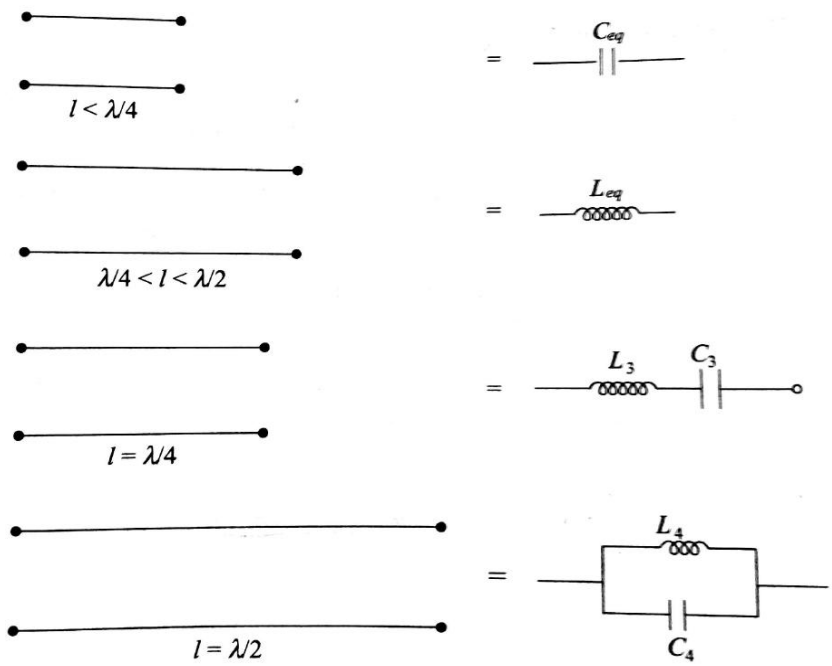


Fig. OC lines at different lengths with equivalent circuits

The input impedance of the short circuited lossless transmission line is given by

$$Z_{sc} = jZ_0 \tan \beta l \quad --- (1)$$

The input impedance of the open circuited lossless transmission line is given by

$$Z_{oc} = -j\, Z_0 \cot \beta l \quad --- (2)$$

The Eq.s(1) and (2) shows that input impedance of an open and short circuited lossless line is a pure reactance. Desired value of the reactance is obtained by varying the electrical length βl of the stubs.

If length of the short circuited line is less than λ/4, It will act as inductance. If length of the short circuited line is greater than λ/4 and less then λ/2, It will act as capacitance. If length of the short circuited line is equal to λ/4, It will act as parallel resonance circuit with high impedance. If length of the short circuited line is equal to λ/2, It will act as series resonance circuit with low impedance.

If length of the open circuited line is less than λ/4, It will act as capacitance. If length of the open circuited line is greater than λ/4 and less then λ/2, It will act as inductance. If length of the open circuited line is equal to λ/4, It will act as series resonance circuit with low impedance. If length of the open circuited line is equal to λ/2, It will act as parallel resonance circuit with high impedance.

4. (b) A transmission line has a characteristic impedance of 300 ohms and terminated in a load Z_L = (150+j 150) ohms. Find VSWR and reflection coefficient.

Ans:

Given that $Z_0 = 300\,\Omega$ and $Z_L = (150 + j150)\,\Omega$

we know that $K = \dfrac{Z_L - Z_0}{Z_L + Z_0}$

$$= \frac{150 + j150 - 300}{150 + j150 + 300}$$

$$= \frac{-150 + j150}{450 + j150}$$

$$|K| = \left| \frac{-150 + j150}{450 + j150} \right|$$

$$= \frac{\sqrt{(-150)^2 + (150)^2}}{\sqrt{(450)^2 + (150)^2}}$$

$$= \frac{\sqrt{150^2 + 150^2}}{\sqrt{450^2 + 150^2}} = \frac{\sqrt{150^2(1+1)}}{\sqrt{150^2(9+1)}}$$

$$= \sqrt{\frac{2}{10}} = \sqrt{\frac{1}{5}} = \sqrt{0.2}$$

$$= 0.447$$

$$\boxed{\therefore |K| = 0.447}$$

we know that VSWR, $S = \dfrac{1 + |K|}{1 - |K|}$

$$= \frac{1 + 0.447}{1 - 0.447}$$

$$= \frac{1.447}{0.553}$$

$$= 2.617$$

$$\boxed{\therefore S = 2.617}$$

Short Answer Questions

UNIT – III

1. Define polarization. List the different types of polarization.

Ans:

The polarization of a uniform plane wave refers to the time-varying behavior of the electric field strength vector at some fixed point in space.
There are three types of polarizations
1. Linear polarization 2. Circular polarization 3. Elliptical polarization

2. State and Express Poynting Theorem.

Ans:

Poynting theorem states that the net power flowing out of a given volume v is equal to the time rate of decrease in the energy stored within v minus the conduction losses.

$$\boldsymbol{P} = \boldsymbol{E} \times \boldsymbol{H} \quad {}^{w}/_{m^2}$$

3. Define the Brewster Angle and critical angle.

Ans:

Brewster angle:
Brewster angle is a particular angle at which no reflection takes place. This occurs when the numerator of reflection coefficient is zero. It exists only for vertically (parallel) polarized waves.

$$\theta_B = \tan^{-1}\sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

Critical angle:
The angle of incident at which the refracted angle becomes 90° is called critical angle and it is denoted by θ_c.

$$\theta_c = \sin^{-1}\sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

4. Find the skin depth in copper if its conductivity σ=5.8X10^7 mho/m at 1 GHz.

Ans:

Given that f = 1 GHz, $\sigma = 5.8 \times 10^7$ mho/m, Assume $\mu_r = 1$

Depth of penetration, δ of an EM wave is given by

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

$$\delta = \frac{1}{\sqrt{\pi f \mu r \mu 0 \sigma}} = \frac{1}{\sqrt{\pi f \mu 0 \sigma}}$$

$$\delta = \frac{1}{\sqrt{\pi x 1 x 10\text{^}9 x 4 \pi x 10\text{^} - 7 x 5.8 x 10\text{^}7}}$$

$$\delta = \frac{1}{\sqrt{\pi\text{^}2 x 23.2 x 10\text{^}9}}$$

$$\delta = \frac{1}{\pi x 0.152 x 10\text{^}6} = 2.094\ \mu m$$

5. List the properties of uniform plane wave.

Ans:

1. A uniform plane wave propagating in z – direction has no z – component of electric and magnetic fields that means $E_z = 0$ and $H_z = 0$.
2. The electric and magnetic fields of EM wave are always perpendicular to each other.
3. The electric and magnetic fields are independent of x and y – directions therefore

$$\frac{\partial^2 \boldsymbol{E}}{\partial x^2} = 0\ and\ \frac{\partial^2 \boldsymbol{H}}{\partial x^2} = 0$$
$$\frac{\partial^2 \boldsymbol{E}}{\partial y^2} = 0\ and\ \frac{\partial^2 \boldsymbol{H}}{\partial y^2} = 0$$

4. The directions of electric and magnetic fields are perpendicular to direction of wave propagation.

UNIT – IV

1. Define the line parameters?

Ans:

The transmission line is described in terms of its line parameters, which are its resistance per unit length R, inductance per unit length L, conductance per unit length G, and capacitance per unit length C. These are also called primary parameters.

2. Define Characteristic impedance.

Ans:

The characteristic impedance Z_0 of the line is the ratio of positively traveling voltage wave to current wave at any point on the line. The characteristic impedance is also defined as the input impedance of an infinite line.

3. What is an infinite line?

Ans:

A line is said to be infinite if all the input signals are consumed by the line and there is no reflected signal.

4. Draw the T Equivalent Circuits.

Ans:

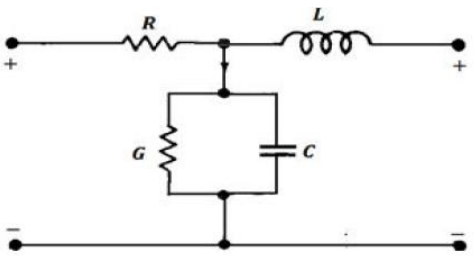


Figure: T-Equivalent circuit of transmission line

5. Mention applications of the transmission lines.

Ans:

Transmission lines are commonly used in power distribution, communications, electrical laboratories and transmission lines such as the twisted-pair and coaxial cables are used in computer networks such as the ethernet and internet.

UNIT – V

1. What is Impedance matching?

Ans:

When the impedance of the source and load of transmission line are matched, then it line is said to be have impedance matching.

2. Define reflection coefficient.

Ans:

The reflection coefficient is defined as the ratio of reflected voltage to incident voltage. The reflection coefficient is also defined as the ratio of reflected current to incident current. It is denoted by K.

3. What is smith chart? Mention its Applications.

Ans:

The Smith chart is the most commonly used graphical techniques in solving transmission problems in simple way. It is basically a graphical indication of the impedance and VSWR of a transmission line as one moves along the line.

Applications of smith chart:

1. Plotting impedance
2. Measurement of VSWR

3. Measurement of reflection coefficient K
4. Measurement of input impedance of the transmission line.
5. Impedance to admittance conversion
6. Finding voltage maximum and voltage minimum locations
7. Stub matching

4. Define the voltage standing wave ratio.

Ans:

The VSWR is defined as the ratio of the maximum voltage to the minimum voltage on the line having standing waves.

$$VSWR = S = \frac{V_{max}}{V_{min}} = \frac{|V_i| + |V_r|}{|V_i| - |V_r|} = \frac{1 + |K|}{1 - |K|}$$

5. Define the stub.

Ans:

A piece of transmission line is called stub. The stub may be open or short circuited. A small section of open or short circuited stub is used as impedance matching device, which can be connected in parallel to the line at a certain distance from the load. This matching device is called stub matching.